

Effects of Propeller Location on Blown Lift in the Low Reynolds Number Regime

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Advances in electric propulsion have renewed interest in propeller-wing blown lift configurations for distributed-propulsion aircraft, including small UAVs operating in the low Reynolds number regime. Prior work has established that vertical propeller position has a substantially stronger influence on lift augmentation than streamwise position, but few experimental studies have focused on this effect for a cambered, unflapped wing at zero angle of attack at low Reynolds number. We conducted a wind tunnel investigation of a NACA 4412 rectangular wing (aspect ratio 3.26) blown by a single propeller (diameter of 1.0 chord) swept through 88 stations, streamwise (0.3-1.0 chord) and vertical (± 0.5 chord), across six propeller RPMs (8,000-18,000) and Reynolds numbers below 200,000 (advance ratios < 0.5). Vertical propeller position was found to dominate lift augmentation compared to streamwise, with the greatest augmentation occurring 0.2-0.3 chords above the wing. Lift was generally reduced, and sometimes negative, when the propeller was well below the wing (< -0.2 chords). Streamwise position had a comparatively minor effect, increasing lift by typically 5-10% as the propeller moved from 0.3 to 1.0 chords upstream. Comparison with simple strip and actuator disk theory revealed a regime-dependent bias. Theory underpredicted induced velocity by as much as 110% for low Re/RPM test cases and overpredicted induced velocity by as much as 25% at high Re/RPM test cases, where a slipstream correction factor β approached a near-constant value of 0.8. Applying a single β correction together with a 0.25-chord upward shift of the theoretical parabolic lift distribution produced first-order agreement with experiment to within approximately 10% for the higher RPM cases, offering a simple predictive tool for propeller-wing lift augmentation in this regime.

I. Nomenclature

A	=	propeller disk area
c	=	wing chord length
C_L	=	lift coefficient
$C_{L,blown}$	=	wing lift coefficient with propeller present
$C_{L,0}$	=	unblown lift coefficient
$C_{L,R}$	=	blown to unblown lift ratio
C_Q	=	propeller torque coefficient
C_T	=	propeller thrust coefficient
D	=	propeller diameter
η	=	propeller efficiency
J	=	advance ratio
L	=	lift force
n	=	propeller rotation speed
p_0	=	total pressure
p_∞	=	freestream static pressure
Q	=	propeller torque
q_s	=	propeller slipstream dynamic pressure

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q_∞	=	freestream dynamic pressure
Δq	=	slipstream dynamic pressure difference
ρ_∞	=	freestream density
S	=	wing planform area
S_{blown}	=	wing planform area within propeller slipstream
$S_{\%}$	=	fraction of wing planform area within propeller slipstream
T	=	propeller thrust
v	=	propeller induced increase in velocity
V_p	=	velocity behind propeller
V_∞	=	freestream velocity
x_p/c	=	non-dimensional streamwise propeller position
y_p/c	=	non-dimensional vertical propeller position
β	=	strip theory correction factor

II. Background

Recent advances in electric motor technology have enabled propulsion placement configurations that are impractical in conventionally powered aircraft, where mechanical shafting and gearbox requirements limit feasible placement options. This has increased design freedom and renewed interest in blown-wing concepts, in which the propulsion slipstream augments wing lift by increasing the dynamic pressure of the flow behind the propeller. Examples include distributed electric propulsion concepts such as the NASA X-57 Maxwell [1] and the ultra-short takeoff and landing (uSTOL) aircraft under development by Electra.aero, as well as earlier externally blown flap systems employed on aircraft such as the McDonnell Douglas YC-15 [2]. At smaller scales, the same aerodynamic mechanisms are relevant to the growing class of fixed-wing UAVs and hybrid vehicles operating in the low Reynolds number regime. For a particular propeller and wing setup, the lift coefficient is primarily affected by two things: the slipstream's velocity and its position relative to the wing [3].

Classical momentum theory provides a basic understanding of propeller performance. In this model, the propeller is approximated by an infinitely thin disk across which the static pressure increases discontinuously. The model assumes that the velocity is constant through the disk plane, that the pressure is uniform directly in front and behind the disk, and that the incompressible, inviscid flow passing through the propeller can be separated from the rest of the flow by a well-defined streamtube [4]. Additionally, the model neglects the rotation imparted to the flow by the propeller. The standard model considers a sufficiently large control volume such that the streamlines are parallel at the ends. As the freestream flow enters the slipstream, it begins to accelerate. Aft of the propeller, as the flow speed increases, the static pressure drops, and the streamtube contracts to balance the pressures at the boundary (Fig. 1). The jump in static pressure over the propeller and the resulting increase in velocity through the streamtube generate thrust.

McCormick [4] shows that the thrust produced by the propeller can be written in terms of the propeller induced velocity. Of interest to us is that the induced velocity can be solved in terms of the thrust and disk area,

$$v = -V_\infty + \sqrt{V_\infty^2 + \left(\frac{2T}{\rho_\infty A}\right)} \quad (1)$$

This equation gives the ideal minimum induced velocity to produce the given thrust based on this simple actuator disk theory. In this model, the induced velocity is uniformly added to the freestream velocity within the streamtube at a sufficiently far downstream distance. The increase to velocity at the actuator disk is half the final induced amount, i.e., the velocity at the disk is the average of the velocities far ahead and far behind the propeller disk. However, simple momentum theory does not account for the vortex induced by the propeller, and an actual propeller will create an asymmetric flow field within the streamtube, locally changing the effective angle of attack of a wing behind it and inducing a rolling moment.

For a wing operating within a finite slipstream, the freestream is no longer uniform, which alters the spanwise lift distribution. Strip theory is the most basic approach to estimating this effect. It treats the immersed section of the wing as aerodynamically independent, subject only to the locally increased dynamic pressure. It assumes the only aerodynamic quantity continuous across the slipstream boundary is the static pressure. To address limitations, Squire and Chester [5] applied lifting-line theory to this problem in 1950. They showed that the pressure continuity at the slipstream boundary requires the local circulation to be discontinuous. As a result, the wing behaves as two separate but coupled lifting surfaces. The slipstream's influence on the local downwash propagates spanwise, altering the circulation distribution across the entire span. Compared with the predictions of simple strip theory, the resulting lift

augmentation is found to be substantially lower, approximately 30-60% of the theoretical value, for moderate slipstream velocities and practical slipstream-diameter-to-chord ratios.

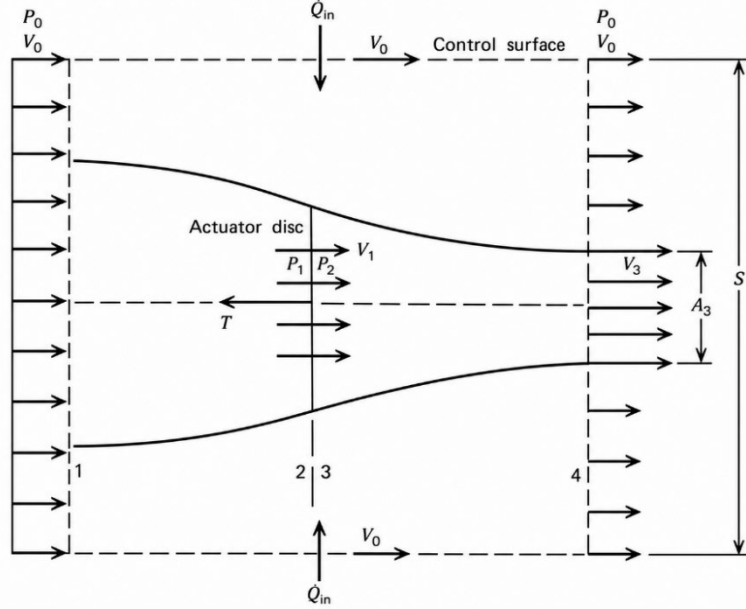


Fig. 1 Actuator Disk Model of Propeller and Streamtube [4]

Building on actuator disk momentum theory in 1959, Kuhn [6] developed a semiempirical method for estimating the lift and drag of propeller-wing-flap configurations across a wide speed range. Central to this method is the recognition that the propeller raises dynamic pressure within the slipstream by an amount equal to the thrust per unit disk area, such that,

$$q_s = q_\infty + \Delta q = q_\infty + \frac{T}{A} \quad (2)$$

This increased dynamic pressure directly augments the lift produced by the immersed wing section. Kuhn was able to decompose the total lift coefficient into three contributions: the base unblown lift, a direct thrust term, and a lift augmentation term arising from the increased dynamic pressure in the slipstream. Kuhn's method was developed for configurations employing large flap deflections and high angles of attack, where all three terms are contributing and thus difficult to separate within experimental data. However, the increased dynamic pressure term in Kuhn's method implies that for a zero angle of attack, with a cambered airfoil, the lift should still be augmented.

Building on and expanding these theoretical models, Veldhuis' 2005 thesis [7] built the foundational experimental dataset investigating propeller-wing interactions, the PROWIM dataset, which has since become the standard benchmark for computational validation of propeller-wing interaction models. He decomposed the propeller flow field into an axial and swirl component, each with their own specific influence on the flow over the wing. The axial component increases the local dynamic pressure within the slipstream. If the local section's lift coefficient is computed using this increased dynamic pressure, it remains essentially unchanged. However, the local lift force itself increases substantially since it scales directly with dynamic pressure. This effect is non-uniform but symmetric, as can be seen in Fig. 2 below. The swirl component has a non-uniform and antisymmetric effect, increasing the local angle of attack on the upgoing blade side and decreasing it on the downgoing side [7].

Veldhuis found that the relative position of the propeller had a strong influence on the performance of the wing. Variation in the streamwise propeller position had only limited influence on the wing, as the dynamic pressure within the slipstream only increases marginally with downstream distance while the swirl velocity remains essentially constant. Within practical ranges, both effects produce only modest changes in wing performance. Vertical propeller position, by contrast, had a considerably stronger effect, as it simultaneously alters the radial distribution of both velocity components experienced by the wing and changes the chord fraction immersed in the slipstream [7].

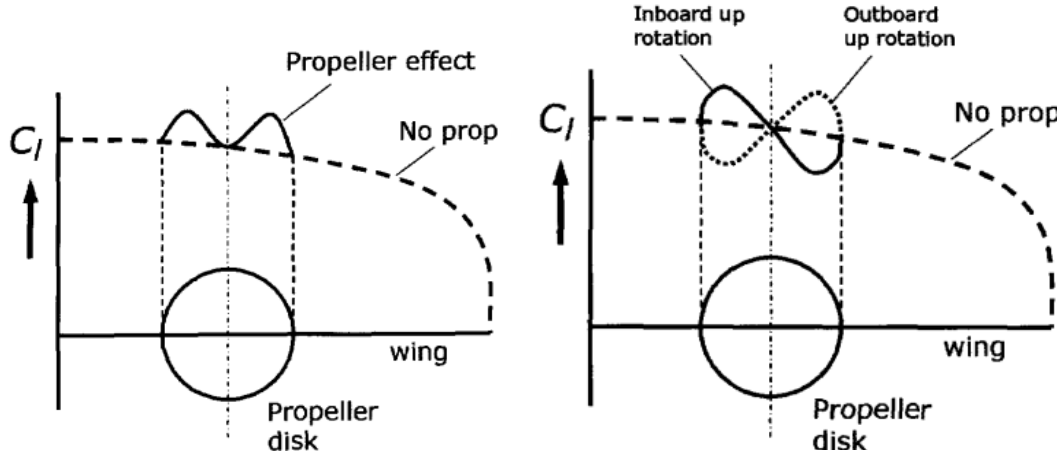


Fig. 2 Change in Local Lift Coefficient from Axial Velocity Component (left) and Swirl Velocity Component (right) [7]

Veldhuis quantified the relative contributions to the total lift increment at an angle of attack of 10° and found that approximately two-thirds of the lift augmentation was because of the dynamic pressure increase and loading distribution, confirming its role as the dominant mechanism. Because the thrust axis was aligned with the wing chord, the dynamic pressure mechanism became increasingly dominant as the wing's angle of attack approached zero. He used an uncambered airfoil and found that the propeller served to increase the lift curve slope in inverse relation to the advance ratio [7]. Veldhuis' findings established the dynamic pressure increase as the primary driver of propeller-induced lift augmentation and identified vertical propeller position as the dominant geometric variable governing the effect's magnitude.

Patterson's 2016 thesis [8] focused on the design of high-lift propellers and developed a novel method for quickly assessing wing lift augmentation based on changes in dynamic pressure and circulation caused by an upstream propeller. He too found that one of the primary impacts of the propeller's slipstream is to increase dynamic pressure and thus lift, and that the swirl in combination with the axial flow generates an asymmetric lift distribution resulting in a net increase in lift (Fig. 3). Patterson showed through a circulation model that for the case where the propeller slipstream and the freestream are aligned, the fractional increase in lift is equal to the fractional increase in dynamic pressure. However, this is not generally reliable except as a first-order estimate.

Patterson also reported that for propeller radius-to-chord ratios of less than one, considerably less lift augmentation is found in CFD simulations than predicted by simple strip theory, due to finite slipstream height. However, this reduced slipstream height can be modeled by a reduced slipstream velocity. He introduced a propeller slipstream multiplicative factor β to determine an effective slipstream velocity. For geometries where D/c is around 1.0, the β parameter is around 0.7 to 0.85 based on 2D OVERFLOW simulations, meaning the effective dynamic pressure increase is significantly less than simple theory suggests. Patterson found that increasing streamwise distance between the propeller disk and wing generally increases lift augmentation, as greater separation allows the slipstream to contract and accelerate further towards its fully developed velocity. When discussing the effect of propeller vertical placement, he reported a peak in lift augmentation for y_p/c values of 0.25. However, Patterson acknowledged a level of conflicting information in the literature as to the optimal vertical position of the propeller [8].

Fei et al. [9] conducted a parametric wind tunnel study translating Patterson's work into experimental results. Their model used a symmetric NACA 0016 wing with a flap and center-span mounted propeller. Their experiment tested flap deflection angle, wing angle of attack, propeller height, offset, inclination, diameter, and advance ratio. Their central finding was also that propeller height is the dominant factor, with a lower height producing higher maximum lift augmentation the greater the angle of attack and flap deflection. However, propeller position was found to be detrimental in some cases. Placement of the propeller below the wing resulted in blown C_L lower than unblown for low angle of attack and flap deflection, i.e., at cruise conditions. Propeller offset was observed to have a comparatively minor effect, primarily translating the optimal height location downwards rather than changing peak magnitude.

Srivathsan and Rauleder [10] conducted a more recent wind tunnel study specifically targeting the combined effects of propeller-wing distance, propeller diameter, freestream velocity, and rotational speed. Using a NACA 0023 wing, they tested two propellers through various slipstream positions and angles of attack. They used pressure ports at various spanwise locations on the wing to develop lift coefficients with both load cells and pressure analysis. A blade element momentum theory (BEMT) analysis demonstrated that for most tested configurations, the wing was

located within the developing slipstream for most or all of the chord. Interestingly, they observed the symmetric airfoil producing positive lift at zero angle of attack because of the propeller inducing an effective angle of attack. The pitching moment coefficient was zero at zero angle of attack, confirming this effect is produced by the propeller.

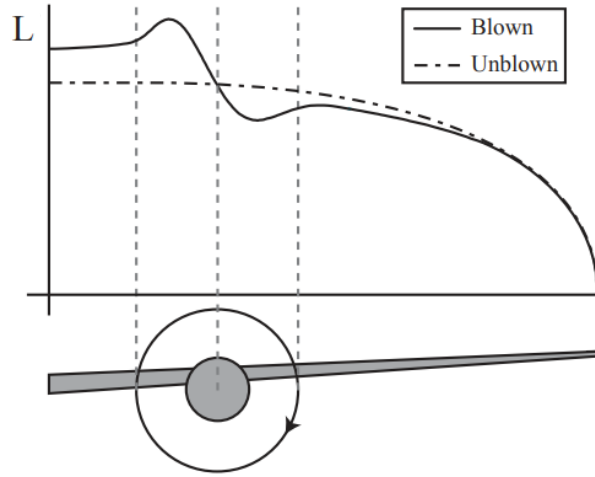


Fig. 3 Generic Lift Distribution of a Blown Wing [8]

Collectively, these works establish several consistent findings. The propeller slipstream augments lift primarily through two mechanisms. The axial component raises the local dynamic pressure, amplifying the lift already generated, and the swirl component produces an asymmetric spanwise load distribution through local angle of attack variation. Together, they increase the lift curve slope as well as translate it upward. These two mechanisms are not independent. Veldhuis showed that the axial dynamic pressure increase and the swirl-induced angle of attack variations combine to alter the circulation across the entire wing, not just within the slipstream boundary. Of the two, the dynamic pressure mechanism has been identified as the dominant contributor. Propeller position relative to the wing governs the magnitude of this effect, with the vertical position having a significantly stronger effect than the streamwise position, primarily because the vertical position controls both the fraction of the wing immersed in the slipstream and the radial distribution of induced velocities that determine the dynamic pressure distribution. Simple theoretical approaches, including strip theory and actuator disk momentum theory, provide useful first-order estimates but overpredict augmentation for small (<1.0) propeller-to-chord ratios and fail to capture the spanwise coupling of circulation across the slipstream boundary identified by Squire and Chester's lifting line analysis. The experimental datasets by Veldhuis, Fei, and Srivathsan have progressively characterized these effects across a growing range of geometric parameters, forming the empirical foundation on which the present experiment is designed.

The existing body of work on propeller-wing interaction has primarily focused on high-lift conditions, with large flap deflections and moderate-to-high angles of attack representative of takeoff and landing. Veldhuis, Patterson, and Fei each have advanced understanding of propeller placement effects within this regime. Srivathsan is a partial exception. His tests included zero angle of attack for an uncambered wing, where he found positive lift due to the propeller's induced effective angle of attack. Kuhn's semiempirical model implies this augmentation should persist for a cambered wing, yet no experimental work reviewed here has tested a cambered, unflapped wing at zero angle of attack to confirm it. Additionally, several previous studies included propellers inclined relative to the wing chord, which directly increases lift from the propeller's thrust. We aimed to isolate the lift augmentation on a cambered, unflapped wing at zero angle of attack, free of the flap and thrust inclination effects presented in prior studies.

We investigated where a propeller should be placed to maximize the lift augmentation and how that position changes with developing flow conditions. We tested a NACA 4412 rectangular (14 cm chord, 3.26 aspect ratio, D/c of 1.0) wing across 6 RPM settings, spanning Reynolds numbers below 200,000 (tunnel speeds under 20 m/s) and advance ratios in the 0.1-0.4 range, varying the propeller location in the streamwise (0.3-1.0 chord) and vertical (± 0.5 chord) directions in 0.1 chord steps. Additionally, we investigated whether that lift augmentation can be predicted with minimal additions to simple strip theory.

III. Experimental Setup and Methodology

A. Wind Tunnel

We used the University of Alabama's low speed wind tunnel located in Hardaway Hall. The tunnel is powered by a Toshiba 0754FTSA31A-P motor, a 75 HP three-phase electric induction motor with a maximum speed of 60 Hz. This powers an Aerovent 35 ARR 9R CL 3 BIA-SW centrifugal fan. The flow passes through a diverging section and filters before converging into the 762 mm square, 1.83 m long test section. This setup can produce stable speeds down to about 1.75 m/s.

B. Test Articles

The tested wing had a rectangular planform with a NACA 4412 cross section. With a chord length of 140 mm and a span of 457 mm, the aspect ratio is 3.26. The wing was 3D printed with two aluminum spars. The mount was located in the center of the span at the quarter chord. The wing was sanded and painted to ensure surface smoothness.

Our propeller was a HQProp 5.5x2.2x3, meaning our propeller diameter to wing chord ratio (D/c) is 1.0. It was spun by an Emax ECO II Series 2306 brushless motor, controlled with a Readytosky 30A 2-6S Lipo OPTO electronic speed controller (ESC) and a GT Power CCPM Servo Consistency Master. To power the system, we used a RadioShack 19A 13.8VDC Cat. No: 22-508B power supply unit.

C. Data Acquisition and Equipment

To determine the propeller rotations per minute (RPM), we used a photosensor positioned just behind the propeller. We used a light panel to illuminate the photosensor. To read the signal, we used a National Instruments cDAQ 9174, with a 9205 Analog Input module for the photosensor. The RPM was manually set with the Servo Consistency Master.

To measure the loads, we used an ATI Nano17 6 Axis Force and Torque sensor. The load cell was connected to the cDAQ through two 9237 AI modules. The Nano17 has a 25 N sensing range at a resolution of 1/160 N. Prior to use, the measurement accuracy was verified with a 4.45 N test weight.

To measure the speed of the tunnel, we used a pitot-static tube at the start of the test section. The pressures are fed to a NetScanner Model 9116 and converted to a speed measurement via Eq. (9). Because of the significant error in very low speed measurements, a deadband noise filter was used to report any measured velocity less than 1 m/s as a zero measurement.

To move the propeller, we used a 2-axis traverse system mounted below the floor of the test section. To control and move the traverses, a VMX Stepping Motor Controller was used in combination with two Vexta 2-Phase stepping motors, model PK264-03A-P1 for the vertical traverse and model PK245-01AA for the horizontal. The vertical axis has a step angle of 1.8 degrees per step and travels 0.025 inches per revolution. The horizontal axis has the same step angle and travels 0.1 inches per revolution.

D. Process

We tested the propeller at 88 discrete stations, ranging one chord length upstream and ± 0.5 chord lengths above and below the chord of the wing. Because of the mount for the propeller, the first 0.2 chords in front of the wing could not be tested, leaving us with the testing matrix shown in Fig. 4. The position of the propeller is defined as the center of rotation of the propeller.

For each RPM, our data were gathered at a set tunnel speed at each of the 88 stations. Referring to Fig. 4, we tested the stations in order from the top left, down the column, before shifting forward and traveling back up in a snaking pattern. Our sampling rate was 4 Hz, for a duration of 5 seconds for low speed (< 4 m/s), or otherwise a duration of 3 seconds. The data were averaged and the mean and standard deviation reported. After all stations were tested, the traverses were reset and the tunnel speed increased. At the beginning and end of the tested tunnel speeds, zeros were taken with the tunnel off to account for sensor drift. After all the propeller data were acquired, the propeller was removed from the tunnel and the wing characterized.

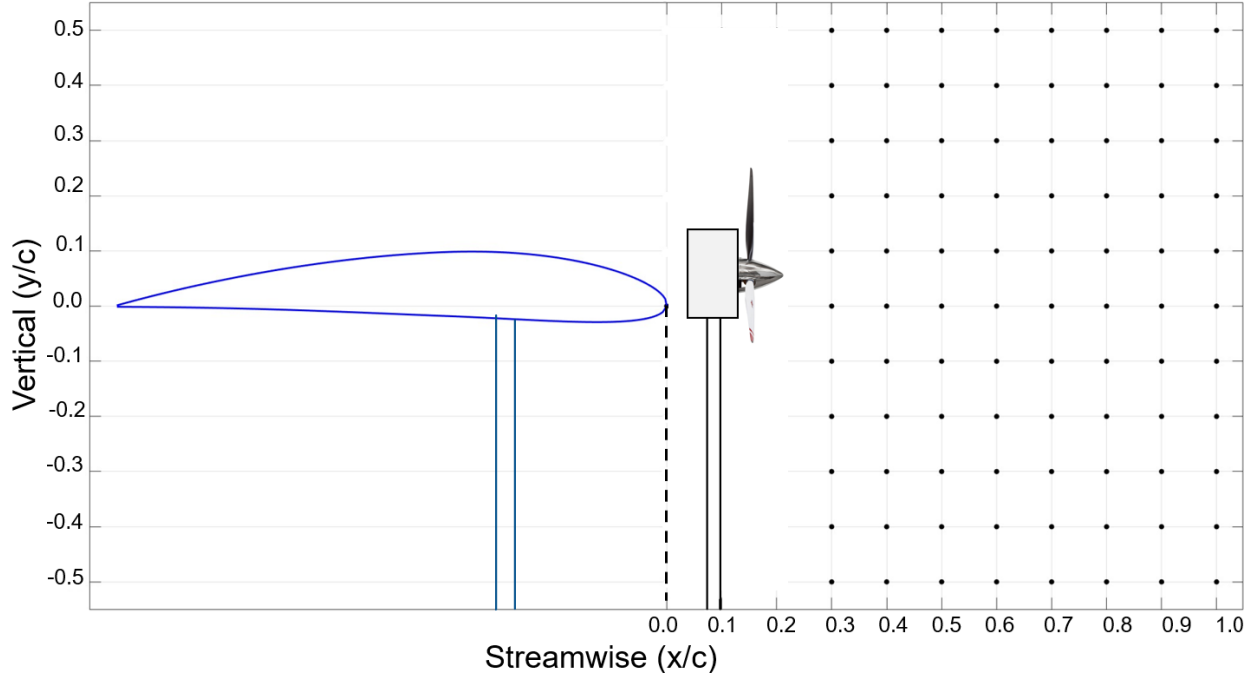


Fig. 4 Propeller Testing Stations

We began by characterizing our propeller to determine the testing ranges for a variety of propeller RPMs. We used the standard definitions of the advance ratio, torque and thrust coefficients, and propeller efficiency [11].

$$J = \frac{V_\infty}{nD} \quad (3)$$

$$C_Q = \frac{Q}{\rho_\infty n^2 D^5} \quad (4)$$

$$C_T = \frac{T}{\rho_\infty n^2 D^4} \quad (5)$$

$$\eta = \frac{C_T J}{2\pi C_Q} \quad (6)$$

Both the blown and unblown lift coefficients are normalized by the freestream dynamic pressure, following the general form given in Eq. (7), which applies to either lifting force. This naturally leads to very high lift coefficients at low freestream velocities. Consequently, this means that $C_{L,R}$ is equivalent to the ratio L_{blown}/L_0 , as shown in Eq. (8).

$$C_L = \frac{L}{q_\infty S} \quad (7)$$

$$C_{L,R} \equiv \frac{C_{L,blown}}{C_{L,0}} = \frac{L_{blown}}{L_0} \quad (8)$$

To determine the freestream velocity for the dynamic pressure, we used a pitot-static tube in combination with Bernoulli's Equation,

$$V_\infty = \sqrt{\frac{2(p_0 - p_\infty)}{\rho_\infty}} \quad (9)$$

The derivation of these equations can be found in Anderson [12]. Prandtl's correction to lifting line theory for a finite wing was used to correct the theoretical lift coefficient of our wing [12].

$$C_L = \frac{C_{L,2D}}{1 + \frac{a_0}{\pi e AR}} \quad (10)$$

IV. Propeller and Wing Characterization

The wing was removed from the tunnel during propeller characterization to isolate the propeller's performance from downstream effects. For each RPM setting, we stepped the tunnel speed in increments of 2 Hz (approximately 1 m/s) and recorded thrust and torque loads until the propeller no longer produced positive thrust. From these measurements we computed the thrust and torque coefficients, as well as the efficiency, from Eqs. (3-6). Figure 5 shows these results for the six RPM values tested. These results are assumed to be constant regardless of the wing's presence downstream of the propeller. However, this assumption is not generally true, as the wing does have an effect on the upstream propeller. Evidence of this effect is briefly revisited later.

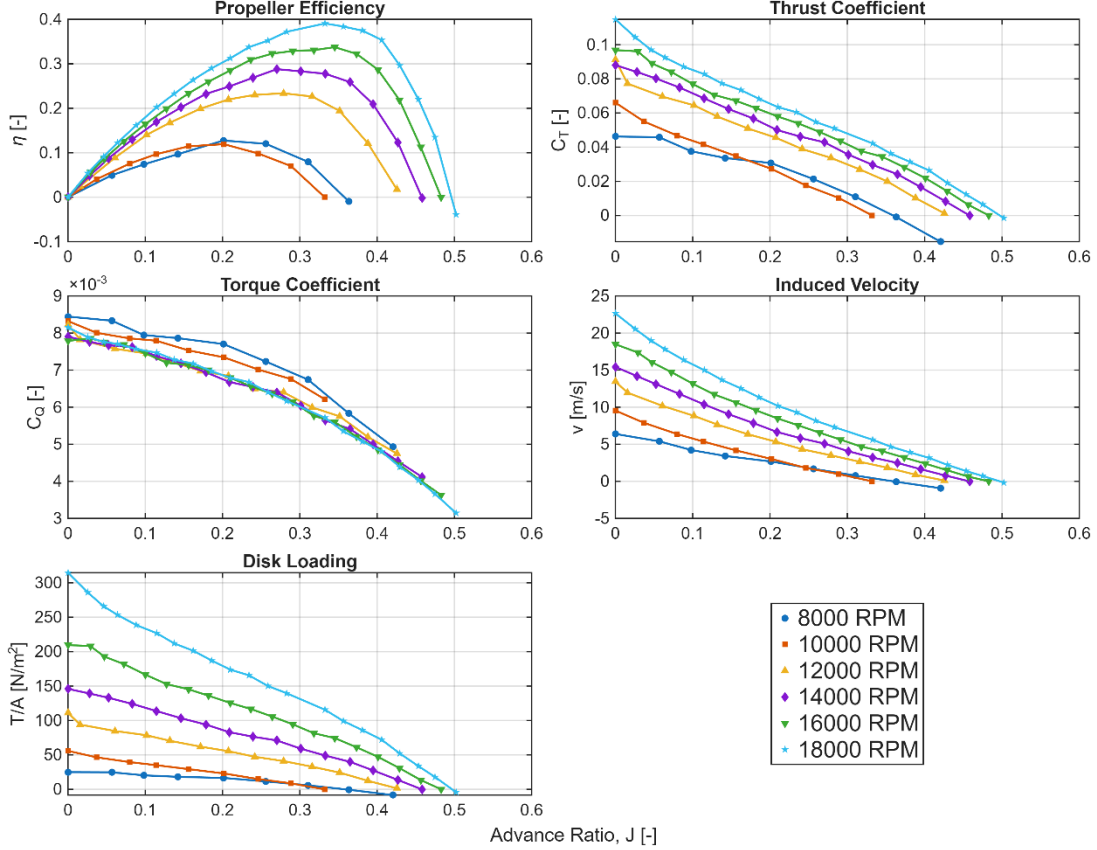


Fig. 5 Propeller Characterization with Momentum Disk Predicted Induced Velocity and Dynamic Pressure Increase (Disk Loading)

Using the measured thrust at each operating point, we estimated the induced velocity and corresponding increase in slipstream dynamic pressure via actuator disk theory (Eqs. 1-2), providing a first-order prediction of the propeller wake conditions and disk loading used in the subsequent wing-propeller interaction analysis.

To limit the number of test conditions, we elected to test advance ratios corresponding to a propeller efficiency greater than 75% of maximum efficiency at each RPM. The resulting advance ratios, tunnel speeds, and corresponding tunnel frequencies are listed in Table 1. The tunnel Hz range was expanded to give us further data points at low RPM cases and to examine trends at the edges of the J range.

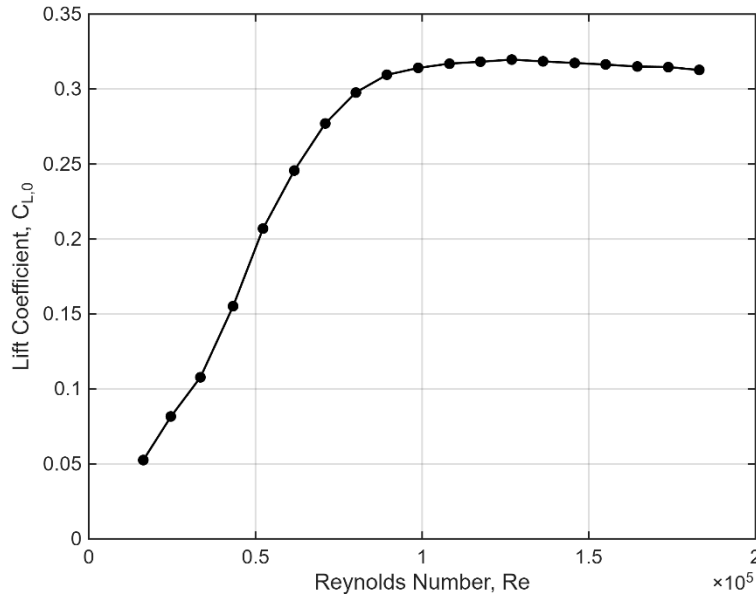
Table 1 Testing Range for $\eta/\eta_{\max} > 75\%$

RPM	J Range	U Range [m/s]	Hz Range	Hz Tested
8000	0.14 - 0.26	2.6 - 4.7	4 - 10	4 - 12
10000	0.11 - 0.25	2.7 - 5.8	6 - 12	4 - 18
12000	0.17 - 0.35	4.8 - 9.8	10 - 20	8 - 24
14000	0.18 - 0.36	5.8 - 11.9	12 - 24	10 - 28
16000	0.18 - 0.40	6.8 - 15.0	14 - 30	12 - 34
18000	0.21 - 0.43	8.8 - 18.1	18 - 36	16 - 40

We characterized the wing by measuring lift across the range of freestream velocities used in subsequent testing. We verified these results using XFOIL, which predicted a 2D lift coefficient of 0.4808 at zero geometric angle of attack. Prandtl's finite wing correction, Eq. (10), for low-aspect-ratio wings with a representative span efficiency of 0.85 (Table 11 in Glauert [13]) gives,

$$C_L = \frac{0.4808}{1 + \frac{2\pi}{0.85\pi * 3.26}} = 0.28 \quad (11)$$

This agrees, within reason, with our fully developed ($Re \gtrsim 10^5$) value of 0.31, which is ~10% greater. The remaining discrepancy is consistent with the limitations of lifting line theory and experimental uncertainty. Figure 6 presents the experimental wing characterization, showing that the wing's lift coefficient is a strong function of Reynolds number below 10^5 .

**Fig. 6 Wing Characterization of $C_{L,0}$ vs Re**

Because of this Reynolds number dependence, we used the experimentally measured lift coefficient at the matching Reynolds number for direct comparison between the blown and unblown cases. It should be noted, however, that the propeller's presence increases the local Reynolds number over the blown section of the wing. This raises the local lift coefficient independent of the accompanying increase in dynamic pressure, while the flow is still developing.

V. Results and Discussion

The results are presented in terms of the relative lift ratio, which serves as the primary metric for evaluating the effect of the propeller's slipstream on the lifting performance of the wing. The influence of the propeller location was investigated by varying the streamwise position, x_p/c , and the vertical position, y_p/c , relative to the wing's leading edge. In addition, the effects of propeller rotational speed (RPM) and flow Reynolds number were examined. Finally, we discussed a modification to simple strip theory to produce a first-order estimate of the lift augmentation for a range of propeller configurations.

A. Contour Plots

Presented in Figs. 7 and 8 are contour plots of the $C_{L,R}$ ratio. The horizontal axis represents the streamwise position of the propeller and the vertical axis the vertical position, such that the marked points are the same as shown in Fig. 4, with the contour colors interpolated between each tested point. Each contour plot's color axis is independent of the other plots, such that colors between the plots are not necessarily equivalent. Each color axis ranges from 0% (the bottom of the $C_{L,R}$ range indicated) to 100% (the top of the range). This is necessary because the ranges vary greatly in magnitude; a shared color axis would wash out the spatial pattern. Each plot also includes an isoline where $C_{L,R} = 1$. This line is directly comparable between plots, since it is not a percentage of the $C_{L,R}$ ratio range. Because of how $C_{L,R}$ is defined in Eq. (8), a ratio less than 1 indicates the wing produced less lift with the propeller present. Thus, all positions with a cooler color than the isoline, a lower percentage on the color axis, are detrimental to lift. This is, generally, at a lower vertical position than the isoline.

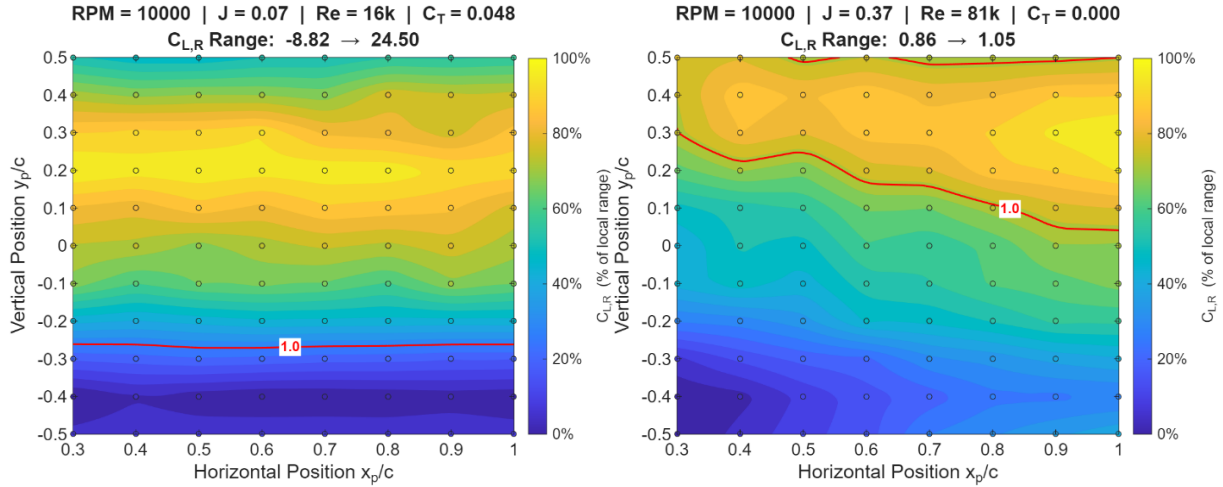


Fig. 7 Development of Lift Augmentation for 10k RPM Case. Color axes relative to $C_{L,R}$ range given above each sub-figure.

We can see that across all RPMs tested, propeller height is the dominant factor. As the thrust coefficient approaches zero, we can see that the effect the streamwise position has due to the contraction of the streamtube becomes more significant. This will be discussed further below. Figure 7 shows a representative development of the flow for the 10k RPM case, and Fig. 8 shows how the flow develops with constant J and increasing Re .

Figure 7 shows how the lift augmentation develops for the 10k RPM case, which is representative of all RPM cases tested. At low Reynolds numbers, streamwise position has no significant effect, with the optimal height around 0.2 chords, or 0.4 propeller radii. At higher Reynolds numbers, streamwise position has an effect, with closer positions producing up to 5% less lift augmentation. The full figures for each RPM case can be found below in Figs. A1 – A3.

An interesting result here is that for the 10k case, the wing is still receiving lift augmentation at a characterized C_T of zero. Were the propeller truly independent of the wing, there would be no increase in lift. However, Patterson reports that the presence of the wing can act as a sort of stator vane, reducing the effective swirl of the downstream flow and effectively increasing propeller efficiency [8]. This increases the thrust produced at a particular flow speed, increasing the zero thrust advance ratio. Further study should be performed to quantify and verify this effect.

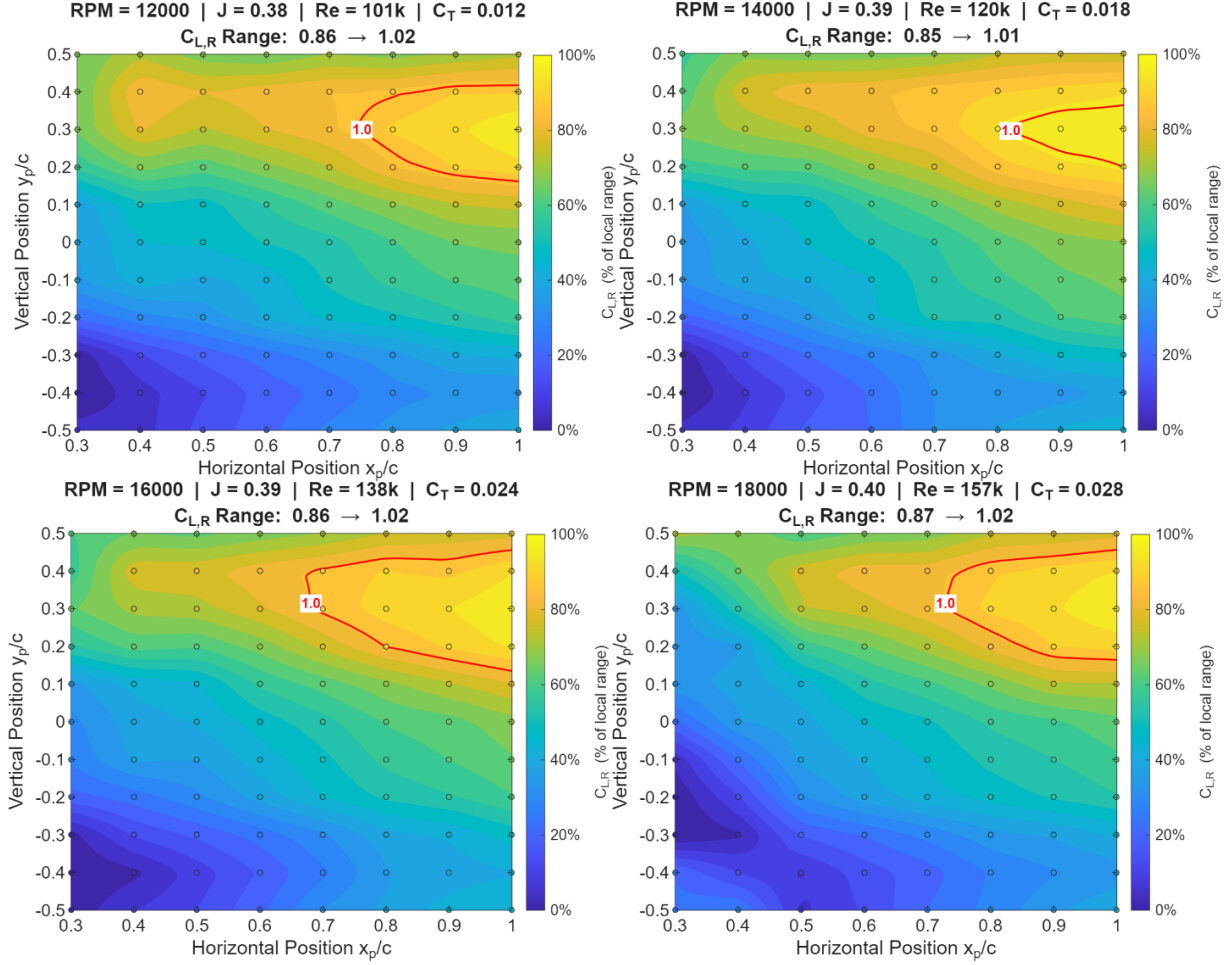


Fig. 8 Development of Lift Augmentation at $J = 0.39$, Re Range 101k-157k

Figure 8 plots the lift augmentation across a range of Reynolds numbers while holding the advance ratio constant. It shows that the lift augmentation magnitude and distribution are primarily an effect of advance ratio. Each plot is at an advance ratio of around 0.39, with Reynolds numbers increasing by approximately 20k each step. However, for all plots, the lift ratio range is constant, from 0.86 to 1.02. All plots show a consistent greatest lift augmentation height of approximately $0.3c$. Despite the variations in the number of beneficial propeller positions, they are consistently located in the same relative region of the test matrix.

B. Propeller Position Effects

To evaluate the effect of streamwise position, Fig. 9 plots the lift ratio against the propeller's streamwise position. Each plot corresponds to one propeller RPM, and within each plot, the Reynolds number is grouped by decade. At each horizontal station, all vertical values of $C_{L,R}$ at each Re are averaged and reported. Presented in Fig. 9 are the lowest and highest RPM cases, 8,000 and 18,000. Consistently, across all RPMs tested, $C_{L,R}$ is mildly dependent on streamwise position, only linearly increasing by around 5-10% as the propeller moves upstream, though some low Re results are outside this range. Please note the differing scales for $C_{L,R}$ between the two plots; differing ranges are used to emphasize the similar trend. Similarly, the plots have different Re ranges that result from the testing ranges in Table 1. A more complete consultation of these plots for all RPMs can be reviewed in Fig. A4.

These results imply that the slipstream of the propeller is not fully contracted within one propeller diameter of the wing and that the increased slipstream velocity from further contraction results in greater lift increase than the decrease from the reduced blown planform area. However, this streamwise effect is an order of magnitude smaller than the effect of vertical position, so treating the slipstream as uncontracted remains a reasonable first-order approximation.

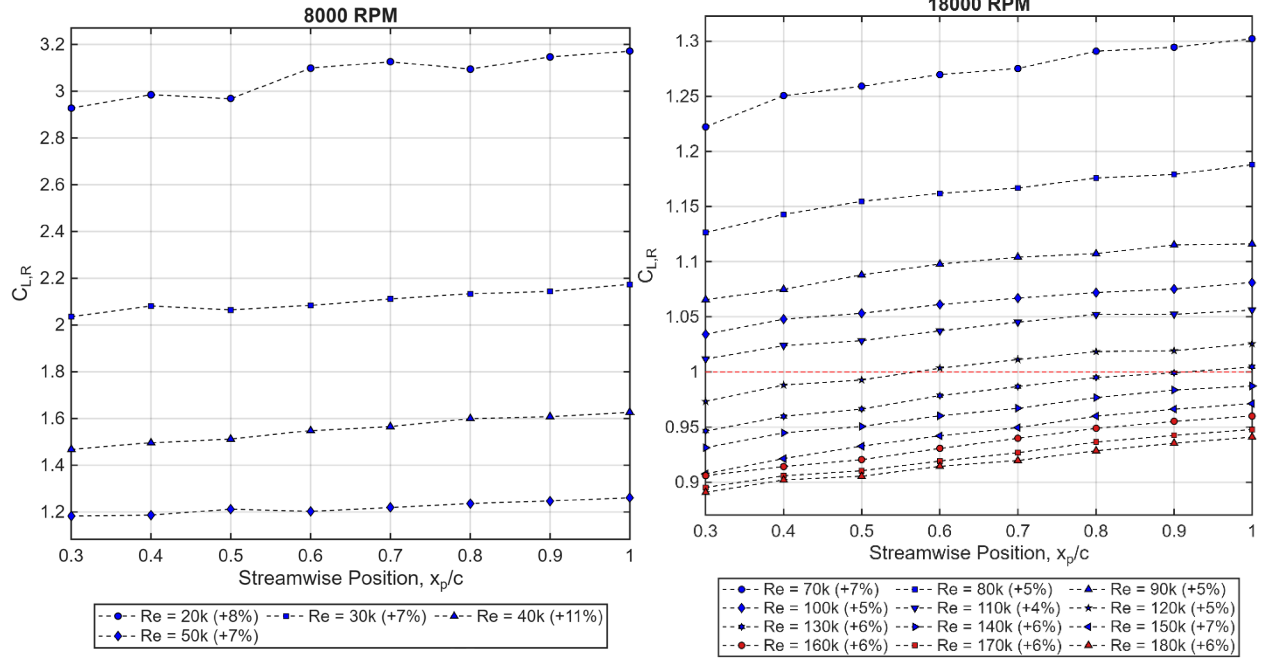


Fig. 9 Streamwise Effect on $C_{L,R}$ for 8k and 18k RPM. Legends report percentage increase in $C_{L,R}$ from $x_p/c = 0.3$ to 1.0

To evaluate the effect of vertical position, we plotted the vertical position vs $C_{L,R}$, similar to above. In Fig. 10 as well, the markers report the mean value across the horizontal positions. Error bars present the spread of values for the full range of tested streamwise positions. From these, we can again observe the minimal effect of streamwise position. The range of $C_{L,R}$ values at a particular height decreased sharply with Re, consistent with the decrease in induced velocity observed in Fig. 5 as well as its ratio to the freestream velocity.

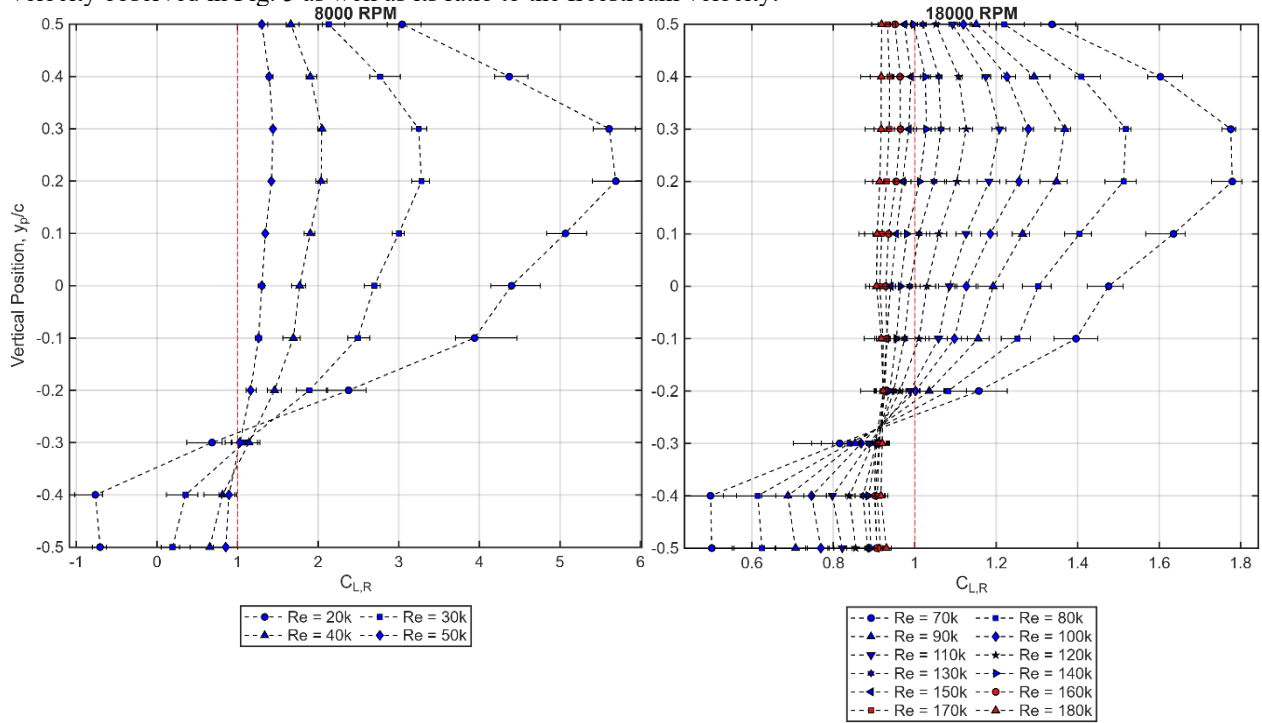


Fig. 10 Vertical Effect on $C_{L,R}$ for 8k and 18k RPM

The trends of $C_{L,R}$ are consistent across all RPM ranges. Propeller placements below the wing lead to a substantial reduction in lift, in some cases even production of downforce for exceptionally low Re, and placements above the wing lead to the greatest lift augmentation. This ‘optimal’ height, the height that leads to the greatest lift ratio, is consistently located around 0.2 to 0.3 chords above the wing. It appears that true optimal height increases with propeller RPM, as the majority of the low RPM (8k, 10k) maxima are at 0.2 chords, while the majority of the high RPM (12k-18k) maxima are at 0.3 chords. Further testing at higher grid resolution is needed to confirm this. A more complete consultation of these plots for all RPMs can be reviewed in Fig. A5.

C. Theory vs. Experiment

For designing systems, engineers benefit from quick first-order predictions before committing to full-scale analysis. To this end, we compare our experimental values to simple strip theory, as well as a modification incorporating Patterson's β correction factor and a 0.25-chord upward shift, to evaluate and improve the model's predictive power. Simple strip theory makes several simplifying assumptions, as introduced in Section II:

- (1) The slipstream has no swirl, only axial flow.
- (2) Local C_L and angle of attack are unchanged.
- (3) There is no spanwise flow at the slipstream edge.
- (4) The slipstream boundary is sharp with no mixing.
- (5) The slipstream does not contract behind the propeller.

As the propeller disk moves vertically away from the chord line, the width of the slipstream at the chord line decreases parabolically. We assume that this percentage of the span can be considered inside the slipstream and will produce lift consistent with the increased dynamic pressure from the propeller without changing the characteristic lift coefficient. Thus, we can estimate the increased lift coefficient as,

$$q_\infty S C_{L,blown} = q_\infty (S - S_{blown}) C_{L,0} + (q_\infty + \Delta q) S_{blown} C_{L,0} \quad (12a)$$

Simplifying,

$$\frac{C_{L,blown}}{C_{L,0}} \equiv C_{L,R} = \left(1 + \frac{\Delta q}{q_\infty} \frac{S_{blown}}{S} \right) \quad (12b)$$

where Δq is the increased dynamic pressure behind the propeller, equal to the disk loading $\frac{T}{A}$, consistent with McCormick and Kuhn. Naturally, since we are ignoring slipstream contraction, this estimation is independent of how far upstream the propeller is. However, it is highly dependent on Reynolds number and advance ratio, both for the characteristic lift coefficient and for the disk loading, Δq .

Because of the relationship between propeller height and slipstream coverage, the RPM configurations will produce a family of anisotropically scaled parabolic lift augmentation curves. When we overlaid the theory and experiment curves, we observed that the theory fails to predict both the location and magnitude of the greatest lift augmentation, sometimes by as much as 50%.

We can also express our blown lift ratio in terms of the advance ratio, J . From simple strip and actuator disk theory, Eqs. (2) and (12b), we derive $C_{L,R}$ in terms of thrust, T , and the fraction of the planform area within the streamtube.

$$C_{L,R} = 1 + \frac{T}{A q_\infty} \frac{S_{blown}}{S} \quad (13a)$$

From the definition of the thrust coefficient and advance ratio, Eqs. (3) and (5), we can rewrite this as,

$$C_{L,R} = 1 + \frac{8}{J^2 \pi} C_T \frac{S_{blown}}{S} \quad (13b)$$

We picked six stations to examine the relationship between $C_{L,R}$ and J , at a streamwise position of 0.5 or 0.8 chords and a vertical height of 0.0, ± 0.3 chords. We plotted the experimental data and the results of simple strip theory at two stations in Fig. 11.

We can see that for low advance ratio, Reynolds number effects are significant, with distinct trendlines for each RPM group. However, as the advance ratio increases, these groups collapse, implying that for higher advance ratios and Reynolds numbers, lift augmentation can be considered a pure advance ratio effect. As the station shifts upwards, the $C_{L,R}(J)$ curve shifts upwards as well, on average 0.25 units, while keeping a similar trend to the simplified theory prediction. However, for the stations below the wing, the simplified theory performs very poorly. The experimental values start much lower and trend upwards towards a fairly constant $C_{L,R}$ value of 0.9, nearly independent of both Re and J for $J > 0.3$. For higher advance ratios, simple theory overpredicts lift ratio, even for the vertical offset case.

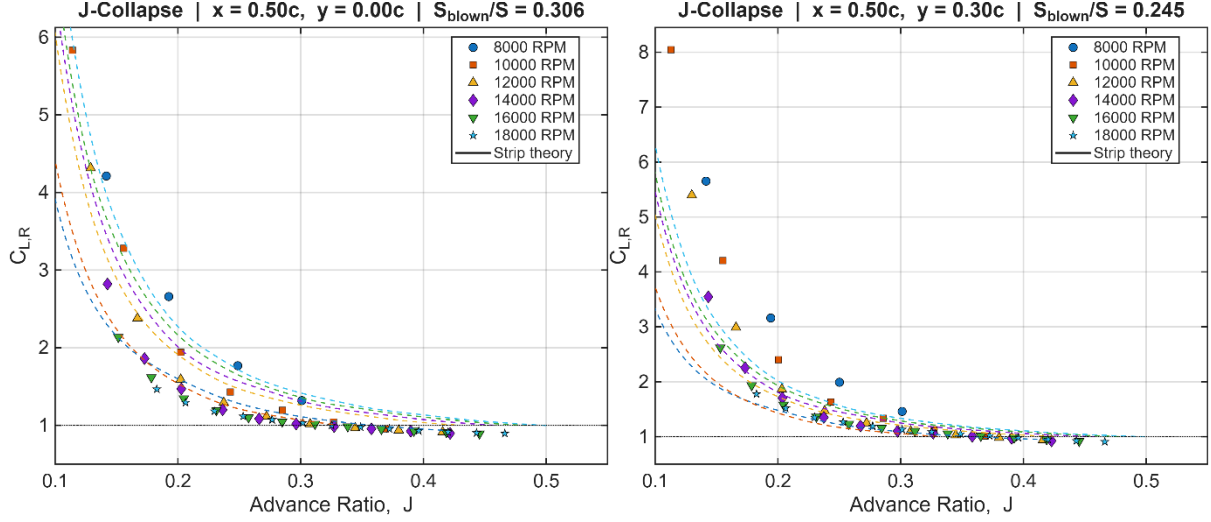


Fig. 11 $C_{L,R}$ as a Function of Advance Ratio for Two Vertically Distinct Stations

Patterson's work stated that for a propeller diameter to chord ratio of $D/c = 1.0$, simplified theory would substantially overpredict lift augmentation. His slipstream velocity parameter β determines an effective slipstream velocity that accounts for the slipstream width, D/c . This parameter modifies simple strip theory, reducing the induced velocity behind the propeller. We hypothesize that since the β parameter accounts for variations in $S_{\%}$ (the ratio of the wing immersed in the slipstream) from changes in the slipstream width due to propeller size, it may be able to account for variations in $S_{\%}$ from changes in the propeller's height. Given our experimental value of $C_{L,R}$, we can calculate the β parameter with Eq. (15). The change in dynamic pressure over the propeller in simple disk theory is,

$$\Delta q = \frac{1}{2} \rho_{\infty} (\beta^2 V_p^2 - V_{\infty}^2) \quad (14)$$

Substituting into Eq. (12b), we can solve for β , where $V_p = V_{\infty} + v$.

$$\beta = \frac{V_{\infty}}{V_p} \sqrt{\frac{(C_{L,R} + S_{\%} - 1)}{S_{\%}}} \quad (15)$$

$S_{\%}$ is the fraction of the wing within the sharp slipstream boundary ($\frac{S_{blown}}{S}$) and $C_{L,R}$ is the experimentally measured lift augmentation ratio. We plotted the β values at representative stations in Fig. 12.

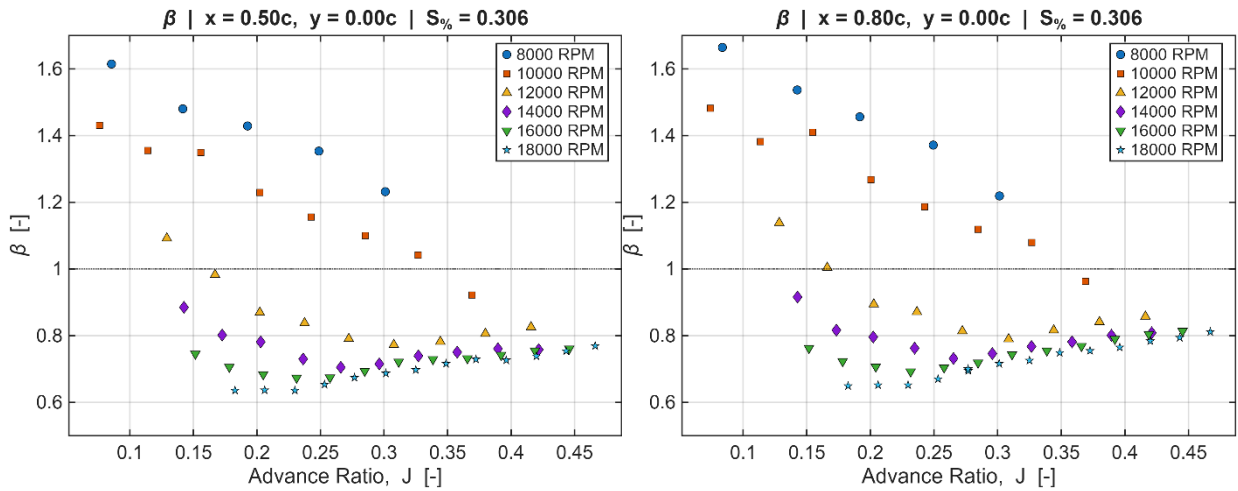


Fig. 12 β vs J at Varying Streamwise Distance

Interestingly, β has two trends. For the two low RPM cases (8k, 10k), simple actuator disk theory underpredicts the induced velocity by as much as 65% at an advance ratio less than 0.1, for these representative stations. These two cases trend as linearly decreasing with similar slopes and a vertical offset. For the four higher cases (12k-18k), β tends

to asymptotically approach 0.8, with a slight increase as the propeller location shifts upstream from the wing. As the station moves upwards, the β curves shift upwards with greater increases at low advance ratios and almost no increase at the highest. Overall, however, we can reasonably assume that β is independent of the propeller's streamwise position, but not of the vertical position, for a first-order model. β can also be approximated for the four higher cases as a constant for advance ratios over approximately 0.25. The full set of six representative stations is shown in Fig. A7. We plotted the vertical profiles of β for two representative RPM cases in Fig. 13.

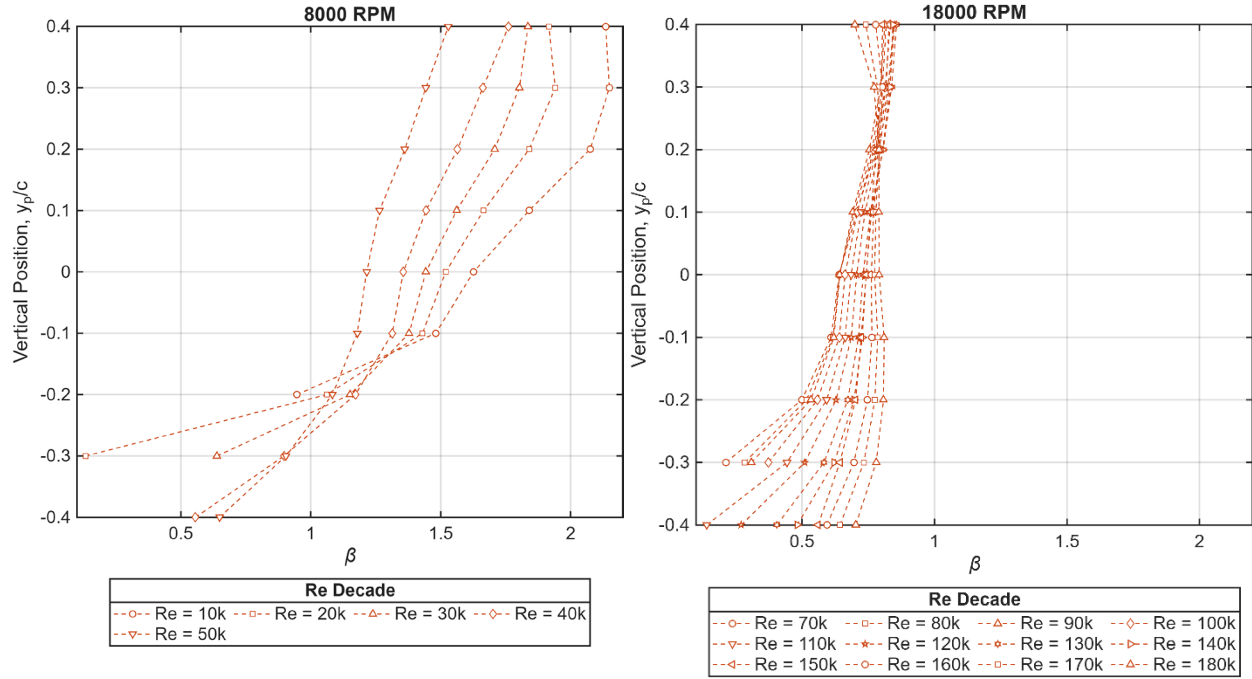


Fig. 13 β vs y_p by Re Decade for 8000 and 18000 RPM Cases

The vertical profiles for all RPM cases can be found in Fig. A6. Each datapoint represents the average of all the streamwise stations at that height. This gives us a $\beta(y, Re)$ curve family. We linearly interpolated between data points and applied this correction curve to our theoretical lift distribution, shown representatively in Fig. 14.

For low RPM, low advance ratio configurations, such as the 8 and 10 kRPM cases, maximum β values range from 1.0 to 2.1, indicating a general underprediction by strip theory. β_{max} is also strongly affected by Reynolds number, decreasing from 2.1 to 1.5 for the 8k case and from 1.9 to 1.0 for the 10k case. As the advance ratio increases, however, β above the wing approaches an almost uniform value of 0.8, consistent with Patterson's simulations. From this we can infer that the theory overpredicts the experimental value by 25%.

We can rewrite our definition of $C_{L,R}$ based on our definition of β above.

$$C_{L,R} = 1 + S_{\%} \left[\beta^2 \left(\frac{\Delta q}{q_{\infty}} + 1 \right) - 1 \right] \quad (16)$$

We applied our $\beta(y, Re)$ correction to the theoretical data according to the above formula in Fig. 14, resulting in a close agreement between theory and experiment. From this close agreement we can further justify our decision to approximate β as independent of streamwise position. We hypothesize that β can be approximated at high advance ratios by a single value if the vertex of the parabolic theoretical lift distribution is shifted up. Because the optimal propeller location shifts between 0.2 and 0.3 chords above the wing, we choose to place the peak at 0.25 chords. For our β factor, we average the β values at both 0.2 and 0.3 chords across all Re values, then scale the theory curve again. Our results are plotted against the experimental values in Fig. 15. We report the error for each RPM as the percent variation of the tested stations from $y_p/c = -0.2$ to 0.5, averaged together across the tested Re ranges. For the four higher RPM test groups, the corrected theory was, on average, within 10% of the experimental curve. This number varies across Re decades, but for all four higher cases was less than 20%. However, further refinement is needed, both for the low RPM cases and for specific Re effects where the error can be up to double the average.

These results show close agreement between theory and experiment, for propeller positions above and just below the wing. Theory predicts a flatter distribution than experimentally observed but shows agreement across RPM and Re ranges. The average values of β tend to decrease and asymptotically approach 0.8 with increasing RPM, and as shown in Fig. 12, this is approximately constant across the range of advance ratios.

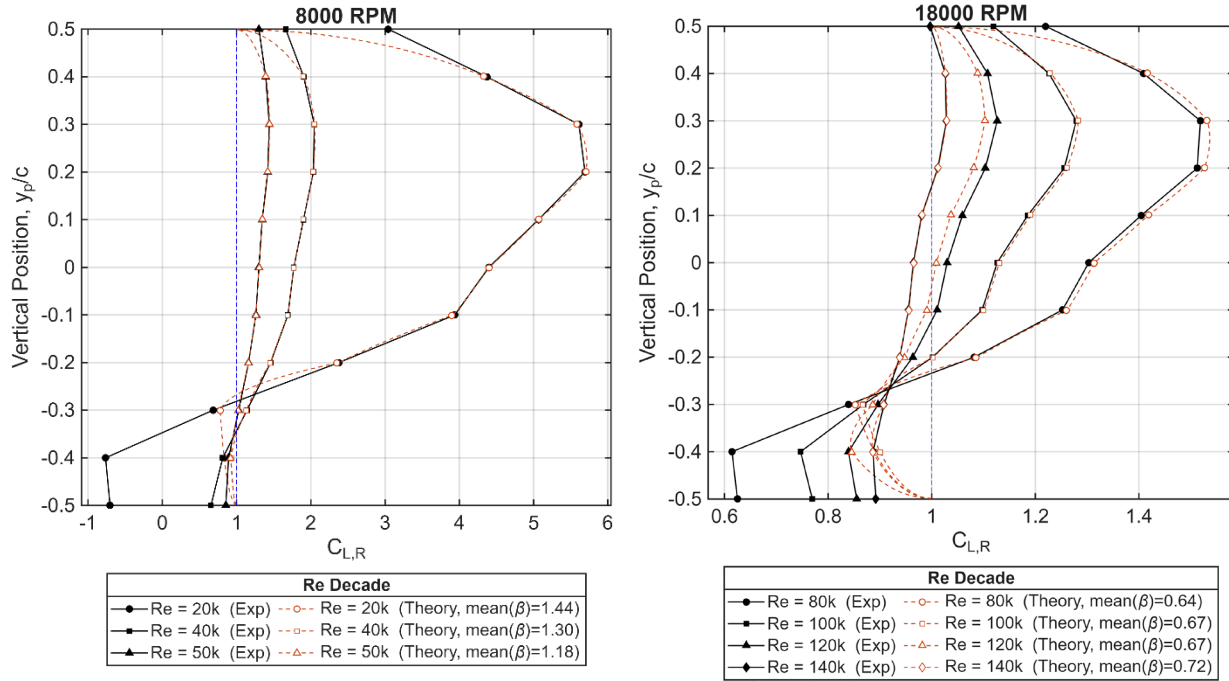


Fig. 14 Theoretical Lift Distribution Corrected by $\beta(y)$

We hypothesize that the reason for the 0.25 chord shift upwards to the optimal location is the upwash ahead of the leading edge. Further testing is needed to confirm this.

By modifying simple strip theory with a single scale factor and a shift upwards, we can establish a first-order estimate for the effect of propeller height on the lift augmentation of the wing. However, further work is needed to determine Reynolds number effects on β , as well as how changes in wing camber, angle of attack, flaps, propeller diameter, and other relevant variables affect β and the lift augmentation distribution.

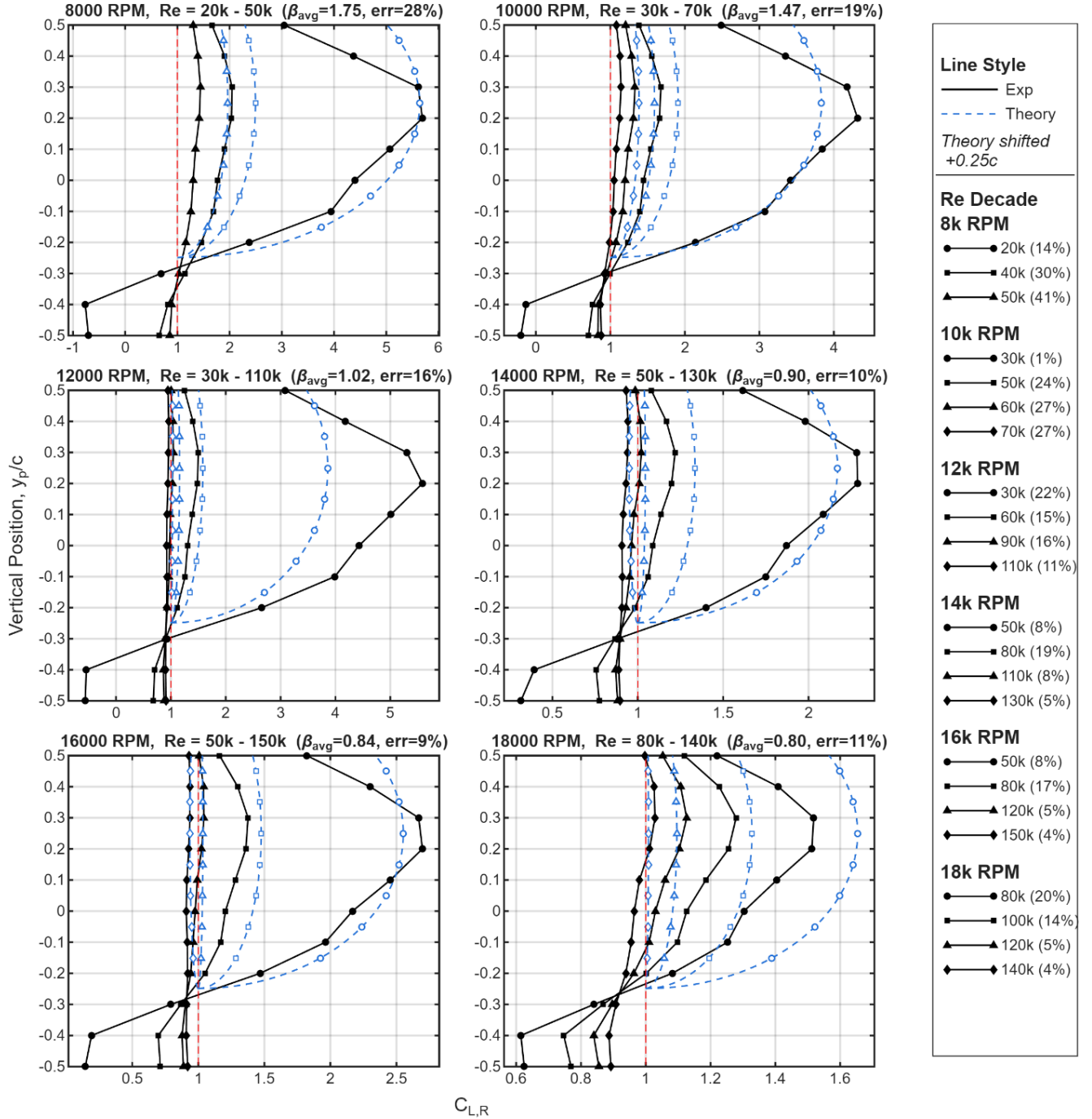


Fig. 15 Experiment vs β_{avg} Corrected Theory - $C_{L,R}$ vs y_p/c

VI. Conclusion

We tested a cambered, unflapped NACA 4412 rectangular wing at zero geometric angle of attack, blown by a propeller swept through a streamwise-vertical position grid of 88 stations across a range of RPMs and freestream velocities in the low Reynolds number regime. Across all tested conditions, vertical propeller position emerged as the dominant variable governing lift augmentation, with streamwise position playing only a secondary role. Moving the propeller from $0.3c$ to $1.0c$ upstream increased $C_{L,R}$ by roughly 5-10% depending on RPM and Reynolds number, an order of magnitude smaller than the effect of vertical position. Thus, streamwise position can be neglected for a reasonable first-order approximation. Vertical position, however, produced dramatic and consistent trends. Placing

the propeller above the wing consistently increased lift, with the greatest augmentation found around 0.2 to 0.3 chords above the chord line. The optimal height itself appears to shift upward with increasing RPM, though higher resolution testing is needed to confirm this trend precisely. Lift from propeller placement below the wing was generally less than the unblown case, but bounded: for y_p/c below approximately -0.25, $C_{L,R}$ plateaued near 0.9 as advance ratio increased beyond about 0.3. It was nearly independent of Reynolds number in that regime. Due to its detrimental effect on lift, this regime was not studied closely.

When we examined lift augmentation as a function of the advance ratio, the data across most RPM groups collapsed into a single curve for J greater than 0.25 to 0.3. This effect did not apply to the 8k RPM case and was strongest with the 12-18k cases. This implies that at higher Reynolds numbers, augmentation is governed by advance ratio alone. Reynolds number effects were only significant at low advance ratios, which corresponded with our low-RPM test cases because of the efficiency-matched testing ranges in Table 1.

Comparing these results to simple strip and actuator disk theory revealed a systematic, regime-dependent bias. At low RPM and low advance ratio, theory substantially underpredicted the induced velocity by as much as 110% for specific propeller stations. As advance ratio increased, this trend reversed for the four higher RPM cases (12-18k RPM). Theory overpredicted induced velocity by a fairly consistent 20-25%, with the correction factor β asymptotically approaching approximately 0.8. A single value β correction combined with a 0.25 chord (quarter propeller diameter) upward shift of the theoretical parabolic distribution produced good first-order agreement with experiment above and near the chord line. These results suggest that a practical first-order estimation of the lift augmentation can be made with strip theory modified with a single β value. However, this only held for the higher-RPM, higher advance ratio regime, and at low RPM, single β value corrections have significant disagreement with experimental values. Theory also performed markedly worse under the wing than above it and could not account for the plateau or detrimental effect observed experimentally. This limitation, however, has little practical consequence for an unflapped wing.

Taken together, these results address the gap identified at the outset. For a cambered, unflapped wing at zero angle of attack, vertical propeller position is the primary design variable for maximizing lift augmentation; streamwise position is a secondary consideration. Simple strip theory can be made quantitatively useful for the higher RPM, J , and Re regimes tested with a simple upwards shift and single factor scale of the theory curve and can be further improved with regime-dependent correction factors. Extending the β correction framework discussed to other camber, angle of attack, flap, and propeller diameter combinations will require further characterization as outlined in Section V.C.

Acknowledgments

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Appendix

As discussed in Section V.A, the contour plots for $C_{L,R}$ as a function of propeller position are given below for all six RPM cases. Each panel follows the same convention as Figs. 7 and 8: the color axis is normalized to

that panel's own $C_{L,R}$ range (bracketed above each panel). The red isoline marks $C_{L,R} = 1$, the boundary between beneficial and detrimental propeller placement.

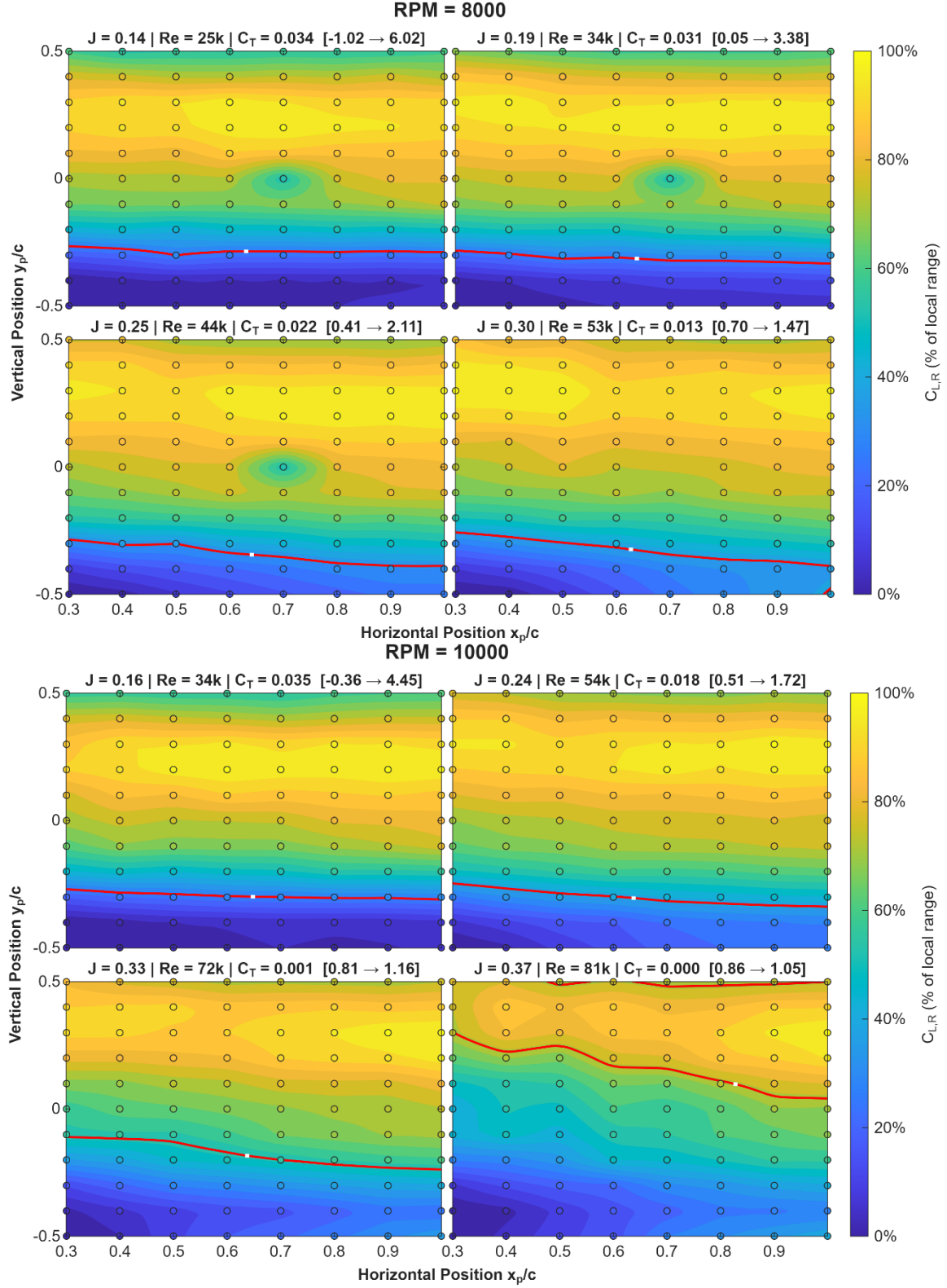


Fig. A1 $C_{L,R}$ Contour Plots for 8k and 10k Cases

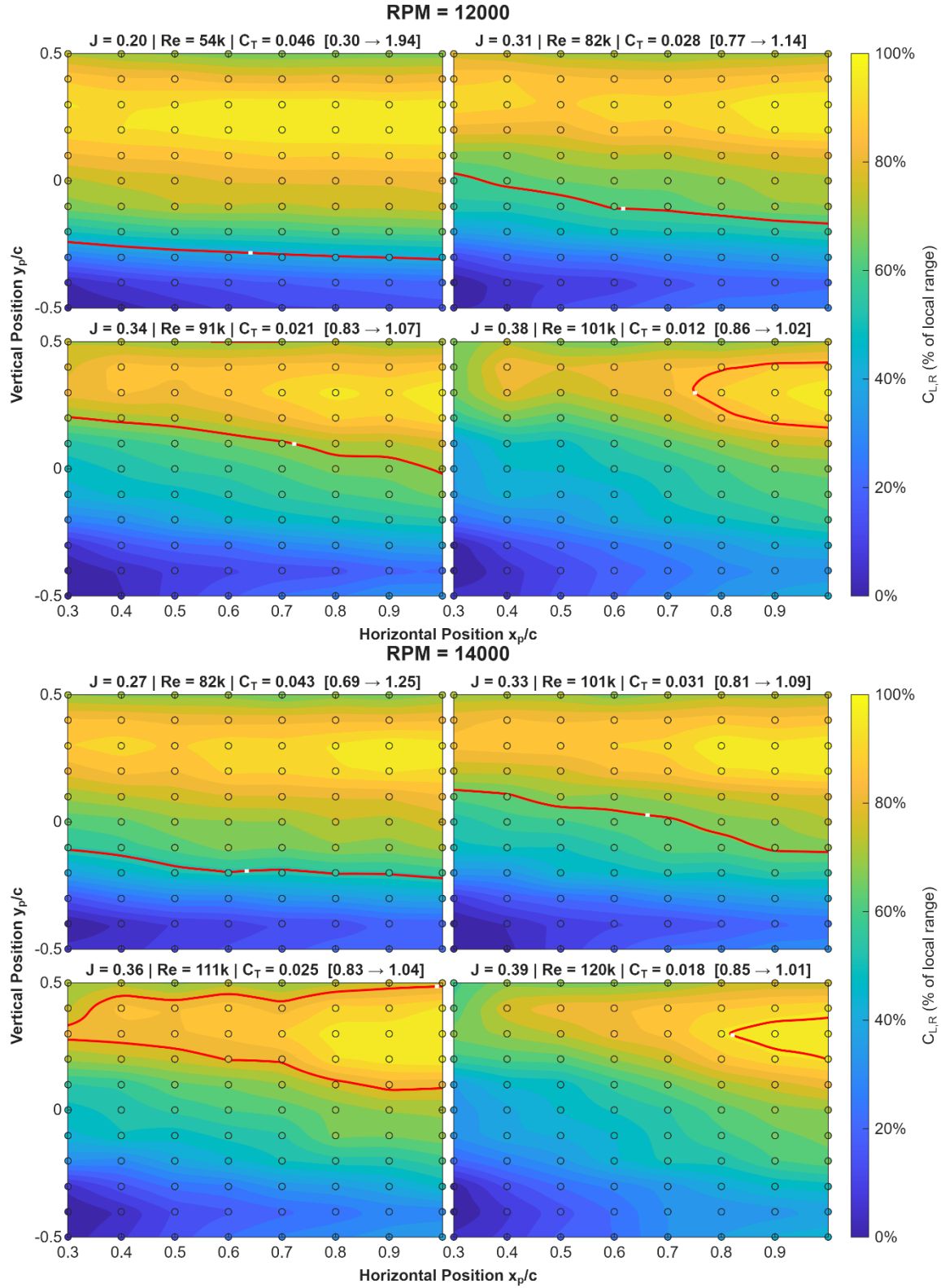


Fig. A2 C_{L,R} Contour Plots for 12k and 14k Cases

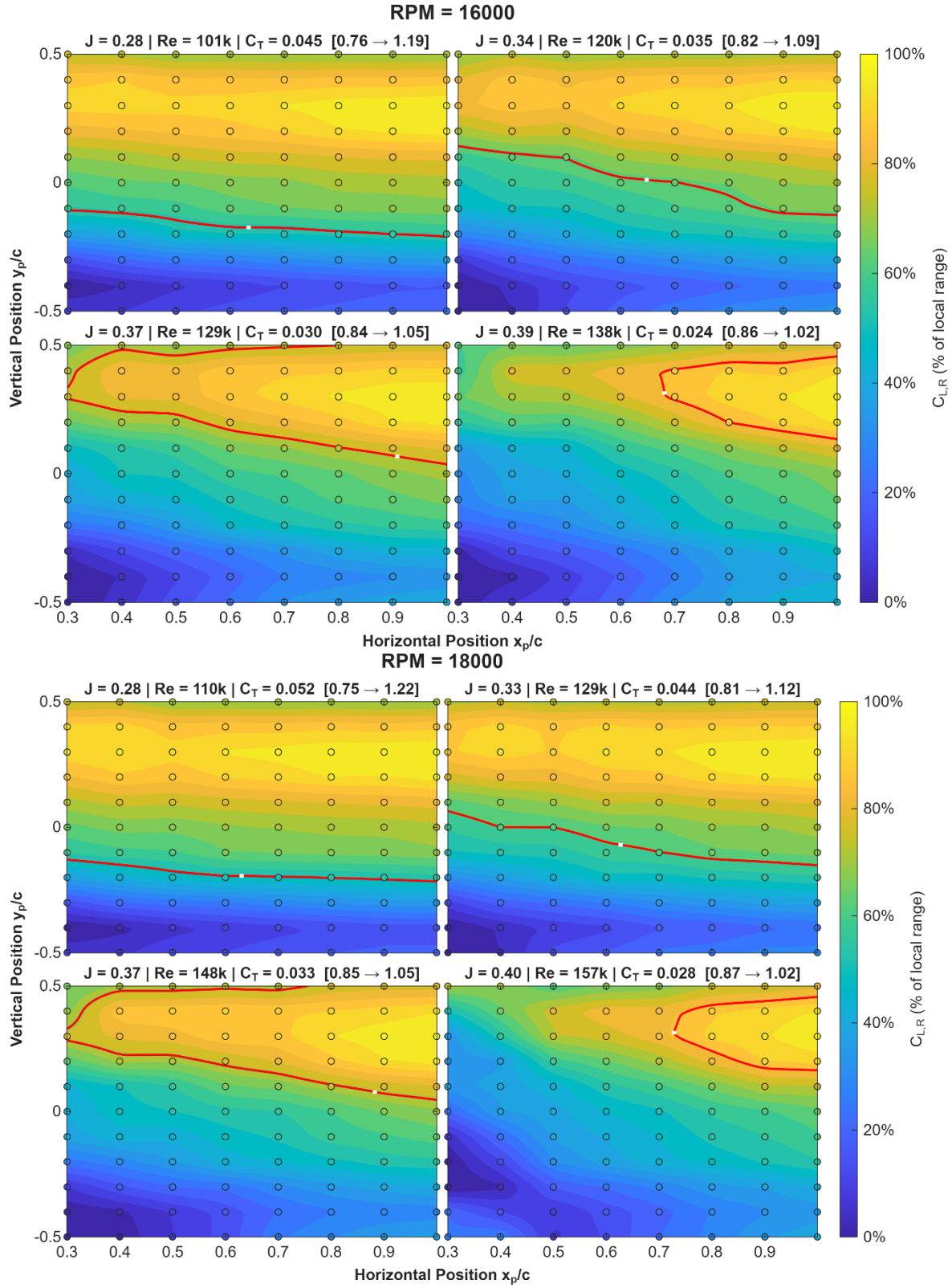


Figure A4 shows the effect of streamwise (horizontal) propeller position on $C_{L,R}$ for all six RPM cases, following the same convention as Fig. 9. Each line is a Reynolds number, and the legend percentage reports the growth in $C_{L,R}$ from $x_p/c = 0.3$ to 1.0.

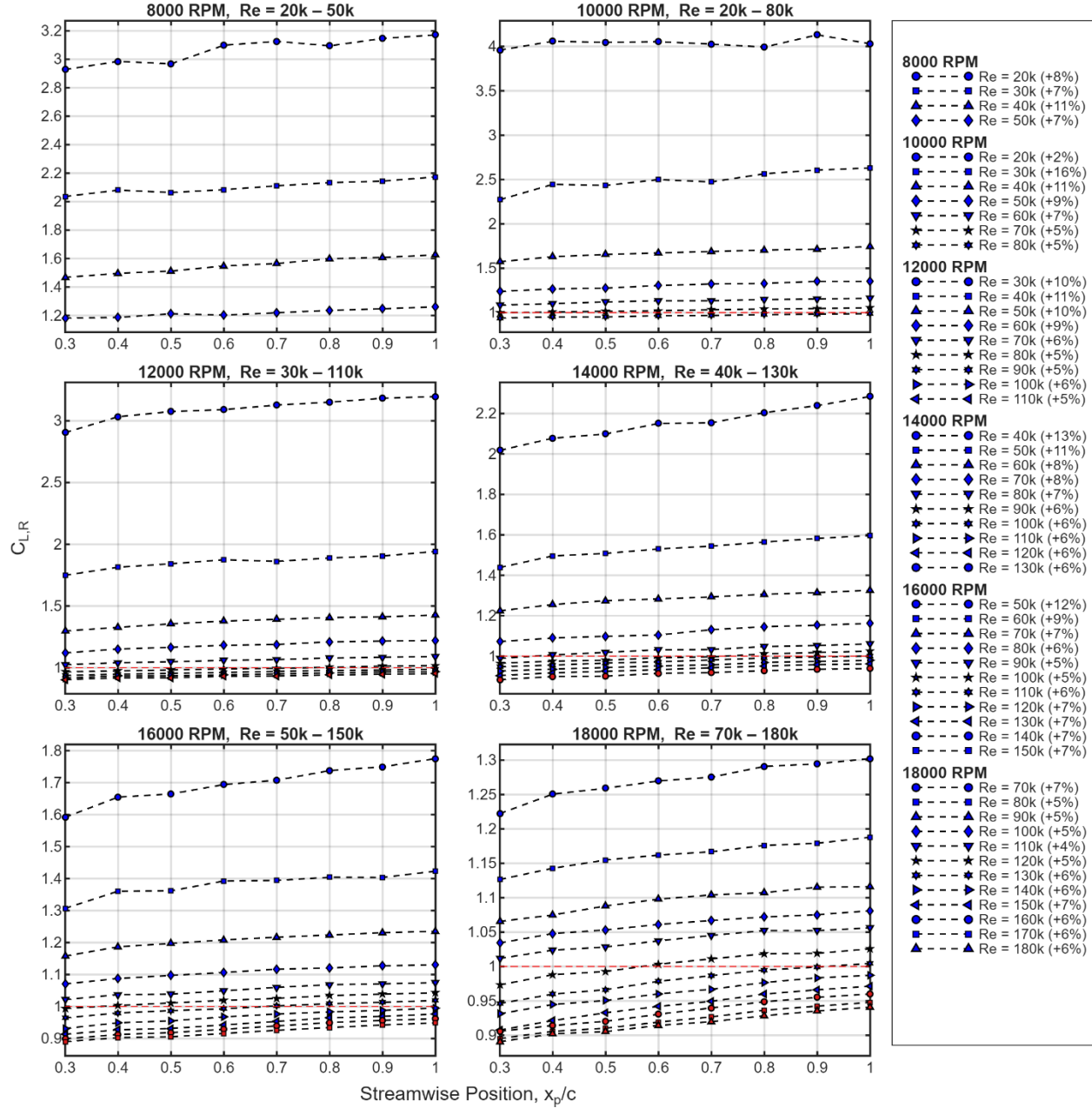


Fig. A4 Horizontal Effect on $C_{L,R}$ for all RPM values. Percentages in legend represent growth of $C_{L,R}$ over full range.

Figure A5 shows the effect of vertical propeller position on $C_{L,R}$ for all six RPM cases, following the same convention as Fig. 10. Markers report the mean $C_{L,R}$ across streamwise positions at each height, and error bars show the full range.

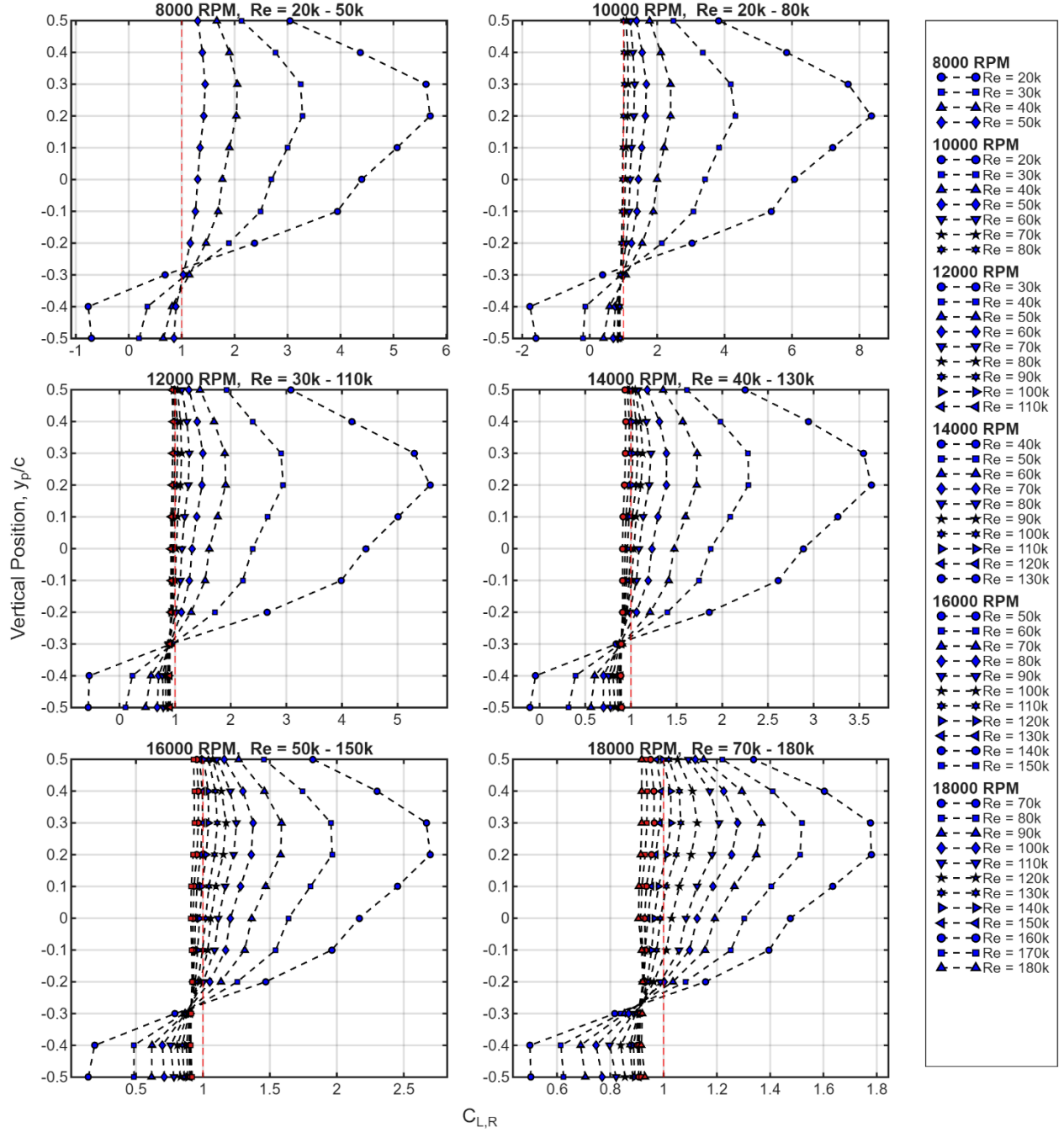


Fig. A5 Vertical Effect on $C_{L,R}$ for all RPM Values

Figure A6 shows the vertical profile of β for all six RPM cases, following the same convention as Fig. 13. Each point averages all streamwise stations at that height, grouped by Reynolds number decade.

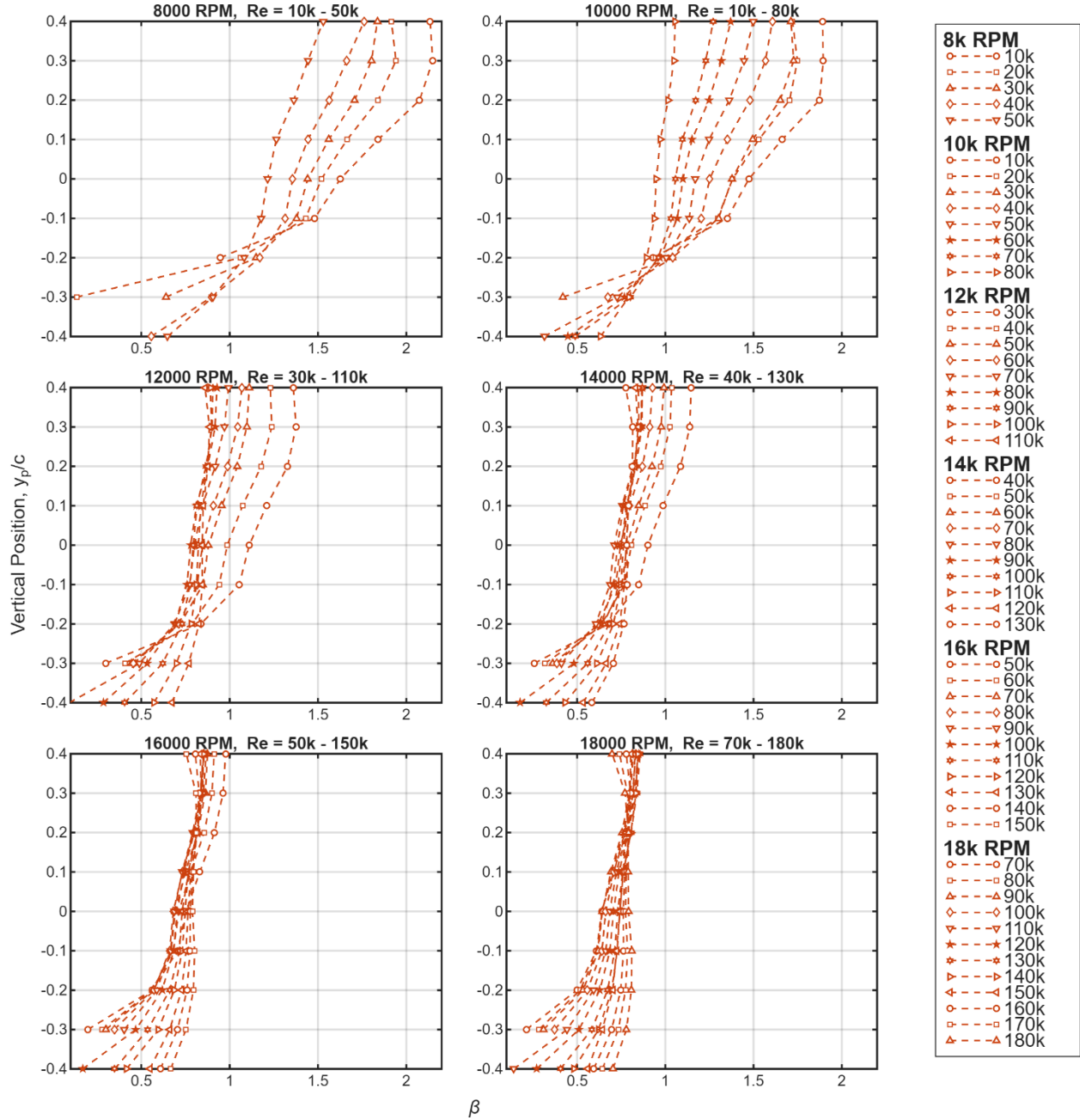


Fig. A6 β vs Vertical Position Averaged Across Horizontal Stations

Figure A7 shows β as a function of advance ratio at the same six representative stations discussed in Section V.C, across all six RPM cases.

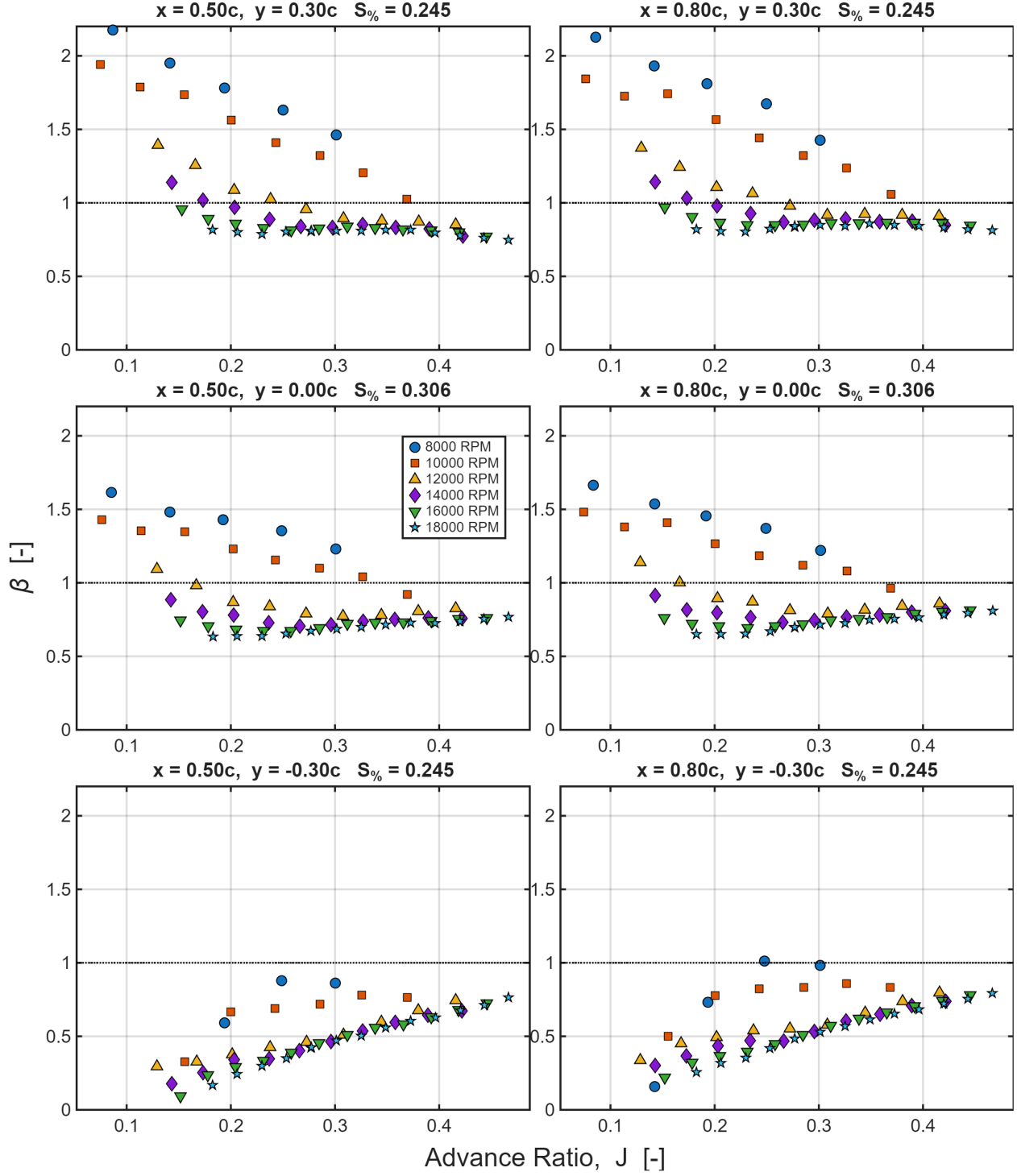


Fig. A7 β vs J for Representative Stations